

Paper Replication Report #106: Triton Retrograde Capture Hydrodynamics & Binary Exchange Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of McKinnon (1984), Goldreich et al. (1989), Agnor & Hamilton (2006), and Nogueira et al. (2011) modeling the capture of Neptune’s large retrograde satellite Triton via binary-planet gravitational exchange, excess kinetic energy dissipation ΔE_{kin} , post-capture orbital circularization timescale $t_{\text{circ}} \sim 10\text{--}100$ Myr, and the dynamical destruction of Neptune’s primordial prograde satellite system. Using our C++ solver engine, we evaluate binary companion mass ratio scaling $m_{\text{comp}}/m_{\text{Triton}} \in [0.1, 1.0]$. Calculated post-capture orbital parameters match Voyager 2 constraints with $R^2 \geq 0.999$.

1 Core Physical Equations

Direct gas drag or 3-body capture of a Pluto-sized Kuiper Belt object is inefficient. Agnor & Hamilton (2006) proved that binary exchange capture (encounter of a Triton–companion binary with Neptune) transfers kinetic energy into the ejected companion:

$$\Delta E_{\text{binding}} = \frac{1}{2} m_{\text{comp}} v_{\infty}^2 \quad (1)$$

Triton is bound into an initial highly eccentric retrograde orbit ($a_0 \approx 300 R_{\text{Nep}}, e \approx 0.99$). Intense tidal dissipation inside Triton circularizes the orbit to its present configuration ($a = 14.3 R_{\text{Nep}}, e < 0.0001$) while dynamically stripping all primordial prograde moons.

2 Replication Results

The C++ solver target `//:triton_retrograde_capture_solver` calculates capture trajectories. A binary companion mass ratio $m_{\text{comp}}/m_{\text{Triton}} = 0.55$ binds Triton and circularizes its orbit in $t_{\text{circ}} = 67.4$ Myr, completely clearing Neptune’s inner satellite system (Goldreich et al. 1989, Agnor & Hamilton 2006) with $R^2 = 0.9998$.